

Jenny (Jia Yi) Pan

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EDUCATION

Columbia University, School of Engineering and Applied Science

Expected May 2027

B.S. in Computer Science

Georgetown University, College of Arts and Sciences

May 2025

B.A. in Political Economy. 3-2 dual-degree program with Columbia University.

Coursework: Distributed Systems, Advanced Security, Parallel Functional Programming, Machine Learning, Advanced Programming, Databases, Stochastic Systems, Linear Algebra, Statistical Computing, Applied Econometrics, Economic Statistics, International Finance.

EXPERIENCE

AI Security Research, Columbia's Software Systems Lab

Summer 2026 - Present

Mentors: Junfeng Yang, Suman Jana. Supported by Google's Cyber NYC Institutional Research Program (summer 2026)

- Engineered Python evaluation software integrating agent harnesses with frontier-model APIs; designed deterministic tests of vulnerable and patched releases to validate reported security failures and maintainer fixes.
- Designed and executed 211 trials of AI red-teaming and jailbreak evaluation, testing prompt injection defenses against supplied-program requests and comparing model responses with execution evidence and task completion.
- Investigate adaptive abstract interpretation and Lean-checked proofs for LLM-designed security analyses, focusing on preserving security-relevant behavior when translating real code into simpler representations.

Financial Engineering Intern, Shanghai Rural Commercial Bank

Summer 2025

- Developed Python and MATLAB workflows for domestic bond-market volatility prediction and tracking; prepared Wind market and macroeconomic data and evaluated volatility models against historical observations.
- Translated research from global banks and funds into concise briefings for the team lead, supporting ongoing market analysis within Wealth and Asset Management.

PROJECTS

TRACE: SQL Injection Detector | Python, Static Analysis, LLM APIs

Spring 2026

Independent project | github.com/jpan301/TRACE

- Designed and implemented a Python analyzer combining static taint analysis, cross-file call graphs and adversarial LLM review to identify SQL injection risks and challenge potential false positives.
- Built a benchmark from real CVE cases and safe production code to compare six configurations. TRACE-Hybrid achieved F1 0.57 versus Bandit 0.20 and a single-pass LLM baseline 0.48 in the project evaluation.

Hermi: Social Outing App | DivHacks - 3rd Place Overall

September 2026

Columbia University | Major League Hacking event | github.com/jackb-22/Hermi

- Led ideation and hand-sketched the end-to-end UX for an iOS app helping students discover local places, plan outings with friends and track in-person exploration.
- Translated user journeys into backend feature and integration requirements for shared plans and check-ins; co-presented the prototype and pitch in a two-person team.

TEACHING

Teaching Assistant, Georgetown McDonough PIVOT Program

2024 - 2025

- Developed statistics, logic and programming instruction for seven students over 20 weeks, adapting explanations and lesson plans to individual learning needs.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, SQL, C, Rust, Haskell, Swift, MATLAB

Engineering: PostgreSQL, Redis, Hono, Zod, Git, Linux, GitHub Actions, Vitest, Playwright

AI & Security: LLM API integration, AI agents, red teaming, jailbreak evaluation, prompt injection testing, static and dynamic analysis, benchmark design

Quantitative: Experimental design, financial data analysis, NumPy, pandas

Product: UI/UX design, user journey mapping, feature requirements, market research